MODULE 04 · DOMAIN 1 - PLAN AND MANAGE AN AZURE AI SOLUTION

Infrastructure and deployment options

Deployment types, data residency, batch and provisioned capacity, version upgrades, agent setup and resiliency.

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A deployment type answers three questions: **where** may inference happen, **how** will you pay, and **how steady** must latency be?

Every deployment type, on three dials

QUESTION	OPTIONS
Where may inference happen?	Global: any Azure region | Data Zone: US, EU or APAC | Standard: one geography
How will you pay?	Standard: per token | Provisioned: reserved PTUs | Batch: half price, asynchronous
How steady must latency be?	Standard types: best effort | Provisioned types: guaranteed throughput

ON THE EXAM

Combine one answer from each row: Data Zone Provisioned is 'EU only, reserved capacity, steady latency'.

Deployment types as hotel bookings

DEPLOYMENT TYPE	THE HOTEL BOOKING
Global Standard	Any hotel in the chain, worldwide. Pay per night
Data Zone Standard	Any hotel in the chain, within one group of countries
Standard	One hotel in one city. Pay per night
Provisioned	A block of rooms reserved in advance. Paid for, used or not
Batch	The overnight laundry service. Half price, back by tomorrow

ON THE EXAM

Spillover, in module 06, is the overflow hotel down the road when your reserved block is full.

Where your data actually goes

- **Data at rest stays in your designated Azure geography**, for every deployment type.
- **Global:** inference may be processed in any Azure region.
- **Data Zone:** inference stays within the US, EU or APAC data zone.
- **Standard and Regional Provisioned:** inference stays within your specified geography.
- The EU zone follows the EU Data Boundary, which can include Norway and Switzerland.

> THE TIEBREAKER
>
> Deployment type only moves inference. A question about where stored data lives is not answered by changing the type.

Start with Global Standard

- **Launches first** when a new model releases, with the **lowest price**.
- **Broadest region coverage** and the **highest default quota**.
- New types arrive in order: Global, then Data Zone, then geography. The last has no guaranteed date.
- At high, steady volume, latency can vary. That is best-effort service.

> ON THE EXAM
>
> Microsoft's rule: move off Global Standard only for a specific reason, such as residency, reserved throughput or batch.

Standard or provisioned?

STANDARD TYPES

Pay for what you use

- Billed per token
- Best-effort service
- Suits variable, bursty traffic
- Latency varies at high, steady volume

PROVISIONED TYPES

Reserve capacity

- Billed for reserved PTUs, used or not
- Guaranteed throughput, steadier latency
- Suits consistent high volume
- Minimum PTUs vary by model

THE SIGNAL

'Consistent high volume' or 'low latency variance' points to provisioned. 'Bursty' or 'unpredictable' points to standard.

Batch deployments

- Send many requests **in one file**, instead of one call at a time.
- **50% cheaper** than Global Standard.
- **Separate quota**, so batch jobs do not slow your online workloads.
- **24-hour target** with no real-time SLA. It can take longer.
- Available as Global Batch or Data Zone Batch. There is no single-region batch.

> ON THE EXAM
>
> 'Overnight', 'weekly', 'not time-sensitive' plus 'reduce cost': that is batch.

Requirement to deployment type

REQUIREMENT	DEPLOYMENT TYPE
No restrictions: newest models, lowest price	Global Standard
Reserved, predictable throughput	Global Provisioned
Keep inference within a data zone	Data Zone Standard
Data zone plus reserved throughput	Data Zone Provisioned
Keep inference within an Azure geography	Standard, or Regional Provisioned
Large asynchronous jobs at lower cost	Global Batch or Data Zone Batch

ON THE EXAM

Not every model supports every type. Always check the model's availability by deployment type and region.

Other deployment options

- **Developer:** for evaluating a fine-tuned model. Deleted after 24 hours, with no SLA or residency guarantee.
- **Managed compute (preview):** open-source or custom models on dedicated GPUs. No deployment types.
- **Instant access (preview):** call supported models by name with no deployment. For trying models.
- **Azure Policy** can block a deployment type, such as Global Standard, across an organisation.

> ON THE EXAM
>
> An organisation that must never process data outside the EU can enforce it with policy, not just with guidance.

Model version upgrade options

OPTION	WHAT HAPPENS
`OnceNewDefaultVersionAvailable`	Upgrades within two weeks of a new default version
`OnceCurrentVersionExpired`	Stays put, then upgrades when the version retires
`NoAutoUpgrade`	Stays put, and stops working when the version retires

ON THE EXAM

Automatic upgrades apply to standard deployment types only. Provisioned and fine-tuned deployments are migrated manually.

Pinning a version for stability is fine. But pinned means **someone owns the upgrade**, or the deployment breaks on retirement day.

Basic or standard agent setup?

BASIC SETUP

Microsoft manages the data

- Conversations and files in Microsoft-managed storage
- Fastest to start
- Recovery options are limited

STANDARD SETUP

You bring your own resources

- Your Cosmos DB, AI Search and Storage
- Data isolated per project, in your subscription
- Supports your own encryption keys
- You own backup and recovery

THE SIGNAL

'Our storage', 'our keys' or 'our backup policy' for agent data means standard setup.

Designing for resilience

- **Foundry has no automatic failover or disaster recovery.** You design it.
- Global and Data Zone Standard spread requests across regions: the simplest resiliency.
- But an outage in your resource's primary region still affects traffic routed there.
- For regional recovery: a second Foundry resource in another region, built from code.
- Put a **gateway**, such as API Management, in front, so failover needs no client change.

> ON THE EXAM
>
> Treat agent definitions, tools and knowledge sources as code, so a standby region can be rebuilt quickly.

Designing infrastructure, step by step

01
Residency
Where may inference happen?

02
Traffic
Bursty, steady, or batch?

03
Type
Pick the deployment type.

04
Versions
Choose an upgrade policy and an owner.

05
Agent data
Basic or standard setup.

06
Resilience
Second region and a gateway, if needed.

ON THE EXAM

Security and networking slot in alongside these steps. Module 07 covers them.

A German insurer is launching a claims assistant. Legal counsel requires that prompts and responses are processed only within the European Union.

Traffic is bursty and hard to predict. The team wants new models as early as possible and a generous default quota.

Which deployment type should the team choose?

A Data Zone Standard, in a region inside the EU data zone.

B Global Standard, because it has the highest default quota.

C Regional Provisioned, in a single region in Germany.

D Data Zone Batch, to keep inference in the EU at lower cost.

◆ PAUSE AND DECIDE

Which deployment type should the team choose?

A Data Zone Standard in the EU.

Inference stays in the EU zone, pay per token for bursty traffic, and earlier models and more quota than geography types.

B Global Standard.

Best quota and newest models, but inference may happen in any Azure region. Fails the legal requirement.

C Regional Provisioned in Germany.

Meets residency, but reserved capacity wastes money on bursty traffic, and new models arrive last.

D Data Zone Batch.

Keeps data in the EU, but a live claims assistant cannot wait up to 24 hours.

THE SIGNAL

One requirement per dial: EU only, bursty, early models. Only one type satisfies all three.

> A retailer must write fresh descriptions for 400,000 products each weekend. The job runs on Friday evening, and results are needed by Monday morning.
>
> The same Foundry resource serves a live customer chatbot that must not slow down. The finance team wants the lowest cost.

How should the team run the description job?

A. A script sending parallel requests to the existing Global Standard deployment.

B. A Regional Provisioned deployment sized to finish the job in a few hours.

C. A model router deployment in Cost mode, called from a scheduled script.

D. A Global Batch deployment, submitting the requests together as a file.

◆ PAUSE AND DECIDE

How should the team run the description job?

A Parallel requests to Global Standard.

Full price, and it competes for the same quota the chatbot relies on.

B Regional Provisioned sized for the job.

Fast, but paying for reserved capacity to run a weekly job is the opposite of lowest cost.

C Model router in Cost mode.

Cheaper per call, but still real-time traffic on shared quota, at no batch discount.

D Global Batch, submitted as a file.

Half the price, separate quota so the chatbot is unaffected, and a 24-hour target fits the weekend.

THE SIGNAL

'Results by Monday' is the permission you need. A 24-hour target inside a 60-hour window is safe.

> A bank's regulatory reporting app uses a Global Standard deployment pinned to one model version, with automatic upgrades turned off, so outputs stay consistent.
>
> One morning, every request starts failing. Investigation shows the pinned model version reached its retirement date overnight.

What practice would have prevented the outage while keeping outputs stable?

A. Set the upgrade option to upgrade as soon as a new default version is available.

B. Keep upgrades off, but name an owner to test and switch before retirement.

C. Move the workload to a provisioned deployment, where model versions are not retired.

D. Recreate the deployment with the same model version under a different name.

◆ PAUSE AND DECIDE

What practice would have prevented the outage while keeping outputs stable?

A Upgrade as soon as a new default arrives.

Prevents the outage, but changes outputs without testing, which breaks the stability requirement.

B Upgrades off, with a named owner who tests and switches in time.

Stability is kept, and the switch happens deliberately before retirement.

C Move to provisioned, where versions never retire.

Provisioned model versions still retire. They just have to be migrated manually.

D Recreate with the same version.

A retired version is not available, whatever the deployment is called.

THE TIEBREAKER

Two requirements pulled against each other: stable outputs and no outage. Only a human process satisfies both.

> A bank is building customer service agents. Conversation history and customer-uploaded files must be stored in the bank's own subscription, encrypted with its own keys.
>
> The bank's resilience team also insists on controlling backup and restore for all of that data.

How should the bank set up its agent infrastructure?

A Basic agent setup, with private endpoints configured on the Foundry resource.

B Basic agent setup, exporting every conversation to the bank's storage each night.

C Standard agent setup, bringing the bank's own Cosmos DB, AI Search and Storage.

D A hub-based project, connected to a storage account in the bank's subscription.

◆ PAUSE AND DECIDE

How should the bank set up its agent infrastructure?

A Basic setup with private endpoints.

Private endpoints change network access, not where agent data is stored or who holds the keys.

B Basic setup with nightly exports.

The primary copy still lives in Microsoft-managed storage, so the requirement is not met.

C Standard setup with the bank's own resources.

Agent data lives in the bank's subscription, with its keys, under its own backup control.

D Hub-based project with bank storage.

The classic setup, not supported in the new portal and not the current agent model.

THE TIEBREAKER

Networking controls how data is reached. Setup type controls where it lives. Do not confuse the two.

Deployment options in five lines

- **Three dials:** where inference happens, how you pay, how steady latency must be.
- **Start with Global Standard.** Data Zone for residency. Provisioned for steady volume.
- **Batch** for work that can wait: half price, separate quota, 24-hour target.
- **Pinned versions break at retirement** unless someone owns the upgrade.
- **Standard agent setup** when you must own the data. **No automatic failover**: design it.

> ON THE EXAM
>
> Next, module 05: bringing Foundry projects into CI/CD pipelines.